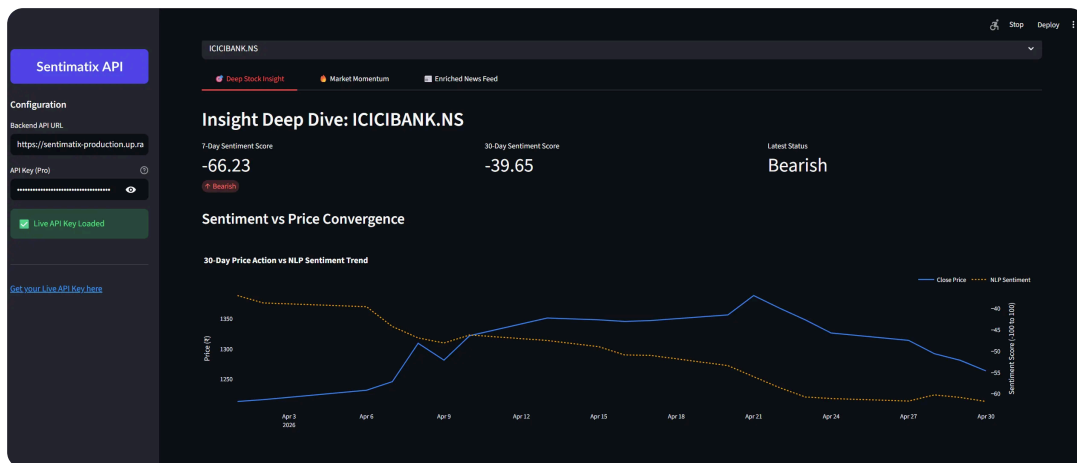


How I built a Real-Time NSE Stock Sentiment Dashboard in 50 Lines of Python



Institutional traders have access to Bloomberg, Refinitiv, and proprietary NLP pipelines. Retail traders in India are flying blind — they rely on headlines and Twitter noise to gauge market sentiment.

I wanted to know how the market *feels* about Reliance before the open. Not what analysts say — what the actual news flow says, right now. So I built this dashboard.

It turns out, you don't need a massive team or millions of dollars to get institutional-grade sentiment analysis. You just need the right APIs and less than 50 lines of Python. Here's exactly how I built a real-time NSE/BSE stock sentiment dashboard.

The Architecture

To keep things lightweight, I used a modern, minimalist stack: - **UI Layer:** Streamlit (Python, zero frontend needed) - **Data/Intelligence Layer:**

Sentimatix API (FinBERT-powered NLP trained on Indian financial news) -

Hosting: Streamlit Community Cloud (free)

The secret sauce here is the **Sentimatix API**. Instead of spending months building custom web scrapers, fine-tuning an LLM, and managing infrastructure to calculate sentiment, I simply use their API. It handles the heavy lifting, delivering real-time sentiment scores for NSE/BSE stocks.

The Core Code

Let me show you how easy it is to pull this data and visualize it.

Fetching sentiment data: First, we make a simple API call to Sentimatix to get the sentiment score for our target stock.

```
import requests

# Fetch 7-day sentiment for Reliance
url = "https://sentimatix-production.up.railway.app/api/v1/sentiment"
params = {"symbols": "RELIANCE.NS", "period": "7d"}
headers = {"Authorization": "Bearer YOUR_FREE_KEY"}

response = requests.get(url, params=params, headers=headers).json()
sentiment_data = response['data'][0]
print(sentiment_data['sentiment_7d']) # e.g. 0.72
```

This endpoint instantly returns a score from -1.0 to 1.0, along with the latest status.

Displaying it in Streamlit: Next, we drop that data into a Streamlit dashboard. With just a few lines, we get a beautiful, interactive metric card.

```
import streamlit as st

st.metric(
    label="Reliance - 7-Day Sentiment",
    value=f"{sentiment_data['sentiment_7d']:.2f}",
    delta=sentiment_data['sentiment_label'],
```

```
        delta_color="normal" if sentiment_data['sentiment_7d'] > 0 else "inver  
)
```

And just like that, we have a functional sentiment dashboard.

The Open Source Handoff

I didn't just build this for myself. I've open-sourced the entire 50-line dashboard on GitHub.

Clone it, tweak the visuals, plug in your own watchlist, or wire it up to a Telegram bot for daily alerts. The UI is entirely yours to customize.

The only thing you need is a free Sentimatix API key to fetch the live data.

Ready to build your own?

The Sentimatix API has a **Free Tier** with enough API calls per day to build, backtest, and explore the market.

👉 [Get your free API key → sentimatix-production.up.railway.app/portal](https://sentimatix-production.up.railway.app/portal)

If you found this useful, drop a ★ on the [GitHub repo](#) — it genuinely helps me keep building open-source tools for retail traders!